Chapter 1 Fields and Galois Theory

1.1 Fields

Definition 1.1.0.1 A **field** is a five-turple $(K, +, \times, 0, 1)$, where:

- $(K, +, 0)$ is an additive group;
- $(K^\times, \times, 0)$ is a multiplicative group;

We have know the construction of

$$Ring \to CRing \to ID \to Fields$$

♠ **Proposition 1.1.0.2** Let A be a domain. Then for every ring morphsim $\phi : A \to K$ where K is a field, there exists factorization

$$\begin{tikzcd} Quot(A) \arrow[r] & K \\ A \arrow[u, hook] \arrow[ur] & \end{tikzcd}$$

○ **Proof** It is well-defined, induced by

$$\frac{a}{b} \mapsto \frac{\phi(a)}{\phi(b)}$$

Universal property omitted. □

Vector Spaces

♢ **Lemma 1.1.0.3** Every $V \in Vect_k$ is free.

○ **Proof** Consider the Bases. (If infinite, rely on AC) □

★ **Corollary 1.1.0.4** Every exact sequence of modules over a field splits.

○ **Proof** Each vector space is free, so is projective. And according to the proposition here: **??** □

Characteristic

Definition 1.1.0.5 Let F be a field, and a ring map

$$\phi : \mathbb{Z} \to F \qquad 1 \mapsto 1$$

Then, the kernel $ker(\phi) = (p)$, $char(F) := p$, is called the **characteristic** of F.

Under this notation:

Definition 1.1.0.6 Let F be a field:

- $char(F) = 0, \mathbb{Q} \subseteq F$
- $char(F) = p, F_p \subseteq F$

is called **prime field**, which is the smallest subfield of F.

1.1.1 Field Extensions

♢ **Lemma 1.1.1.1**
If F is a field and R is a nonzero ring, then, any ring homomorphism

$$\phi : F \to R$$

is injective.

○ **Proof** Suppose $a \in Ker(\varphi); a \neq 0$. Then, notice that

$$\varphi(1) = \varphi(aa^{-1}) = 0 \Rightarrow 1 = \varphi(1) = 0$$

Contradiction. □

Definition 1.1.1.2 If F is a field contained in a field E, then E is said to be a **field extension** of F. We shall wirte

$$E/F$$

to indicate that E is an extension of F.

Category of Field Extension

With 1.1.1.1, we have known: The ring morphsims between fields must be injective. And: Those morphisms represent the field extensions.

For morphism $\phi : F \to K$, K can be seen as F-algebra.

Definition 1.1.1.3 Let k be a field. There is a **category of field extensions** of k.

- Objects: Morphisms $k \to E$;
- Morphsims: $\begin{tikzcd} E \arrow[rr] & & E' \\ & k \arrow[ul] \arrow[ur] & \end{tikzcd}$

  Take the notation of morphsims $Mor_k(E, E')$.

Definition 1.1.1.4 (Tower of Extensions) A **tower of fields** $E_n/E_{n-1}/\dots/E_0$ consists of a sequence of extensions of fields $E_n/E_{n-1},\dots,E_1/E_0$.

Definition 1.1.1.5

♢ **Lemma 1.1.1.6** ...

◦ **Proof** ... □

Finite Extensions

Definition 1.1.1.7 Let F/E be an extension of fields. The dimension of F considered as an E-vector space is called **degree of extension** and is denoted as:

$$[E:F] := dim_E(F)$$

Algebraic Extension

Definition 1.1.1.8 Consider a field extension F/E. An element $\alpha \in F$ is said to be **algebraic** over E is α is the root of some nonzero polynomial with coefficients in E.

If all elements of F are algebraic then F is said to be an **algebraic extension** of E.

Minimal Polynomials

Algebraic Closure

1.1.2 Separation Extension

Chapter 2 The Fundamental Theorem of Galois Theory

Chapter 3 Algebraic Closure

Chapter 4 Infinite Galois Extensions